

**Engineering Economics
(MG3008)**

Final Exam

Date: June 2nd 2025
Course Instructor(s)
Mr. Syed Haris Mujtaba

Total Time (Hrs): 3
Total Marks: 115
Total Questions: 5

221-6272
Roll No

EE-6 C
Section

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Student Signature

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Attempt all the questions.

Solve everything on the answer sheet and show each step.

In those question where the table is provided fill the values in them and return the question paper with answer sheet.

CLO #: 2 Interpret the concept of time value of money

Q1: By applying time value of money determine the future worth in year 10 of a lease that yielded income of \$14,000 in years 1 through 4 and then amounts that decreased by 5% per year through year 10. Use an interest rate of 18% per year. [20 marks]

CLO #: 3 Analyze cost effectiveness of alternatives using engineering economy methods

Q2: A company is analyzing two alternatives for improving the exterior appearance of the building. The building can be completely painted at a cost of \$6500. The paint is expected to remain attractive for 4 years, at which time repainting will be necessary. Every time the building is repainted, the cost will be 20% higher than the previous time. Alternatively, the building can be sandblasted now and every 6 years at a cost 40% greater than the previous time. If the company's MARR is 10% per year, what is the maximum amount that could be spent now on the sandblasting alternative that would render the two alternatives indifferent over a study period of 12 years? [20 marks]

CLO #: 3 Analyze cost effectiveness of alternatives using engineering economy methods

Q3: A piece of equipment has a first cost of \$150,000, a maximum useful life of 7 years, and a market (salvage) value described by the relation $S = 120,000 - 20,000k$, where k is the number of years since it was purchased. The salvage value cannot go below zero. The AOC series is estimated using $AOC = 60,000 + 10,000k$. The interest rate is 15% per year. Compute the economic service life. [20 marks]

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CLO #: 4 Develop cash flow sequences that include the effects of taxes and depreciation
Q4-a: Construct a loan repayment table for an asset worth \$950,000, interest is 12% per year for 5 years; when equal end of year payments are made. [10marks]

N	Big-Bal	Interest	Payment	Principal	Remaining	Acc-Principal
0	950,000	0	0	0	950,000	0
1	800,461	114,000	263,539	149,539	800,461	149,539
2	632,977	96,055	263,539	167,484	632,977	317,023
3	445,395	75,457	263,539	187,882	445,395	504,605
4	235,303	53,447	263,539	210,692	235,303	714,697
5	0	28,236	263,539	235,303	0	950,000

Q4-b: The Marginal tax rates are given below, compose all the remaining columns of the table: [5 Marks]

Range	Tax Rate	Limit-From	Limit-Up-to	Maximum Tax of Range	Maximum Tax Incurred
150,000	8%	0	150,000	12,000	12,000
200,000	12%	150,001	350,000	24,000	36,000
200,000	18%	350,001	550,000	36,000	72,000
150,000	24%	550,001	700,000	36,000	108,000
2500,00	30%	700,001	950,000	75,000	183,000
Above	40%	950,001	Above	Above	Above

CLO #: 4 Develop cash flow sequences that include the effects of taxes and depreciation
Q5-a: Create a MACRS table with 150% DB also calculate the depreciation amount and book value for each year of an asset, bought for \$900,000 and has a life of 6 years. [20 marks]

N	Bal	Dep1	Remain	N*	Straight	Dep	Currency	BV
1	900	12.5	87.5			12.5	112,500	787,500
2	87.5	21.875	65.625	5.5	15.909	21.875	196,875	590,625
3	65.625	16.406	49.219	4.5	14.583	16.406	147,654	442,971
4	49.219	12.304	36.915	3.5	14.062	14.062	126,558	316,413
5	36.915	9.228	27.687	2.5	14.062	14.062	126,558	189,855
6	27.687	6.921	20.766	1.5	14.062	14.062	126,558	63,279
7	20.766	20.766	0	0.5	7.031	7.031	63,279	0

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Q5-b: Carry Forward from Q4 and Q5(a): From information calculated in table for interest in Q4-a; also use the MACRS table of Q5-a, and make the rest of the income statement using the tax table in Q4-b, with Revenue, Material, Labor and Overhead and their relative inflations given below: (20marks)
Assemble the complete table of income statement.

		0	1	2	3	4	5
Revenue	18	1,300,000	1534 000	1810 120	2135 941	2520 410	2974 066
Material	17	350,000	409 500	479 115	560 564	655 851	767 355
Labor	16	280,000	328 800	376 768	439 050	506 978	586 094
Overhead	16	150,000	174 000	201 840	234 184	271 595	315 050
RMC			625 700	752 397	904 193	1085 978	1303 584
Dep			112 500	196 875	197 654	126 558	63 279
EBIT			513 200	555 522	756 581	959 420	1240 305
Interest			114 000	960 55	759 67	634 47	282 36
EBT			399 200	459 467	680 582	905 973	1212 069
Tax			448 56	55704	103 337	169 791	200000
EAT			354344	403763	577 243	736 182	1012 069

(20)